



ICT372 MOBILE COMPUTING T226

All information in the Subject Outline is correct at the time of approval. KOI reserves the right to make changes to the Subject Outline if they become necessary. Any changes require the approval of the KOI Academic Board and will be formally advised to those students who may be affected by email and via Moodle.

Information contained within this Subject Outline applies to students enrolled in the trimester as indicated

1. General Information

1.1 Administrative Details

Associated HE Award(s)	Duration	Level	Subject Coordinator
Bachelor of Information Technology (BIT)	1 trimester	Level 3	Imtiazuddin Ali Khan MOHAMMED imtiazuddin.mohammed@koi.edu.au P: +61 (2) 9283 3583 L: 7-11, 11 York St. Consultation: via Moodle or by appointment.

1.2 Core / Elective

Elective subject for the BIT

1.3 Subject Weighting

Indicated below is the weighting of this subject and the total course points.

Subject Credit Points	Total Course Credit Points
4	BIT (96 Credit Points)

1.4 Student Workload

Indicated below is the expected student workload per week for this subject

No. Timetabled Hours/Week*	No. Personal Study Hours/Week**	Total Workload Hours/Week***
4 hours/week (2 hour Lecture + 2 hour Tutorial)	6 hours/week	10 hours/week

* Total time spent per week at lectures and tutorials

** Total time students are expected to spend per week in studying, completing assignments, etc.

*** Combination of timetable hours and personal study.

1.5 Mode of Delivery Classes will be face-to-face or hybrid. Certain classes will be online (e.g., special arrangements).

1.6 Pre-requisites ICT272 Web Design and Development and ICT205 Cyber Security

1.7 General Study and Resource Requirements

- Dedicated computer laboratories are available for student use. Normally, tutorial classes are conducted in the computer laboratories.
- Students are expected to attend classes with the requisite textbook and must read specific chapters prior to each tutorial. This will allow them to actively take part in discussions. Students



should have elementary skills in both word processing and electronic spreadsheet software, such as Office 365 or MS Word and MS Excel.

- Computers and WIFI facilities are extensively available for student use throughout KOI. Students are encouraged to make use of the campus Library for reference materials.
- Students will require access to the internet and email. Where students use their own computers, they should have internet access. KOI will provide access to required software.

Resource requirements specific to this subject: The resource requirements for this subject are Office 365. Microsoft Azure and cross -platform development tools in Visual Studio Online, Xamarin, Mobile Device Management (MDM), Windows Virtual Machine.

1.8 Academic Advising

Academic advising is available to students throughout teaching periods including the exam weeks. As well as requesting help during scheduled class times, students have the following options:

- Consultation times: A list of consultation hours is provided on the homepage of Moodle where appointments can be booked.
- Subject coordinator: Subject coordinators are available for contact via email. The email address of the subject coordinator is provided at the top of this subject outline.
- Academic staff: Lecturers and Tutors provide their contact details in Moodle for the specific subject. In most cases, this will be via email. Some subjects may also provide a discussion forum where questions can be raised.
- Head of Program: The Head of Program is available to all students in the program if they need advice about their studies and KOI procedures.
- Vice President (Academic): The Vice President (Academic) will assist students to resolve complex issues (but may refer students to the relevant lecturers for detailed academic advice).

2 Academic Details

2.1 Overview of the Subject

This subject focuses on mobile computing using mobile development techniques and environments. Students will learn to develop applications for current and emerging mobile computing devices, performing tasks at all stages of the application development life cycle from inception through to implementation and testing. Topics covered include app life cycle; small device programming; platform; web applications for mobile; cross platform development. Students then put these skills to use on an application idea of their choosing, and individually develop their idea into a complete, medium-sized mobile application.

2.2 Graduate Attributes for Undergraduate Courses

Graduates of Bachelor courses from King's Own Institute (KOI) will achieve the graduate attributes expected under the Australian Qualifications Framework (2nd edition, January 2013). Graduates at this level will be able to apply a broad and coherent body of knowledge from their major area of study in a range of contexts for professional practice or scholarship and as a pathway for further learning.

King's Own Institute's generic graduate attributes for a bachelor's level degree are summarised below:

	KOI Bachelor Degree Graduate Attributes	Detailed Description
	Knowledge	Current, comprehensive, and coherent and connected knowledge

	Critical Thinking	Critical thinking and creative skills to analyse and synthesise information and evaluate new problems
	Communication	Communication skills for effective reading, writing, listening and presenting in varied modes and contexts and for transferring knowledge and skills to a variety of audiences
	Information Literacy	Information and technological skills for accessing, evaluating, managing and using information professionally
	Problem Solving Skills	Skills to apply logical and creative thinking to solve problems and evaluate solutions
	Ethical and Cultural Sensitivity	Appreciation of ethical principles, cultural sensitivity and social responsibility, both personally and professionally
	Teamwork	Leadership and teamwork skills to collaborate, inspire colleagues and manage responsibly with positive results
	Professional Skills	Professional skills to exercise judgement in planning, problem solving and decision making

Across the course, these skills are developed progressively at three levels:

- **Level 1 Foundation** – Students learn the basic skills, theories and techniques of the subject and apply them in basic, standalone contexts.
- **Level 2 Intermediate** – Students further develop the skills, theories and techniques of the subject and apply them in more complex contexts, and begin to integrate this application with other subjects.
- **Level 3 Advanced** – Students demonstrate an ability to plan, research and apply the skills, theories and techniques of the subject in complex situations, integrating the subject content with a range of other subject disciplines within the context of the course.

2.3 Subject Learning Outcomes

This is a Level 3 subject.

On successful completion of this subject, students should be able to:

Subject Learning Outcomes	Contribution to Graduate Attributes
a) Identify problems and understand current mobile technologies, their strengths and their limitations.	  
b) Apply current software technologies, framework architecture and standards used in mobile application development.	     
c) Design and implement mobile applications using either Java or C languages including the use of standard Application Development Interfaces (API).	     
d) Demonstrate an understanding of security, ethics, and privacy in mobile application development,	    



with an emphasis on frameworks like GDPR and Australia's APPs.

2.4 Subject Content and Structure

Below are details of the subject content and how it is structured, including specific topics covered in lectures and tutorials. Reading refers to the text unless otherwise indicated.

Weekly Planner:

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 1 29 June	Introduction to mobile app development, creating virtual device, Testing mobile apps	Chs.1-5	Tutorial exercises based on lecture topics are intended to stimulate discussion. Students are required to understand how to set up an Android Studio Development Environment by creating an example Android app.
Week 2 06 July	Creating user interface, Mobile app Architecture	Chs.6-9	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on Code compilation, code folding, and code reforming. Summative worth 2%
Week 3 13 July	Anatomy of Android Application, Android View Binding, Activity Lifecycle Personal, Legal, Ethical, and Social Issues	Chs.10-12	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on code compilation, code folding, code reforming. The application of the ACS Code of Professional Conduct through practical case study scenarios Summative worth 2%
Week 4 20 July	Restoring the state of activity, views, groups and layouts, creating user interface. Data Credentials and Privacy	Chs.13-16 & 18-25	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on visible process, activity stack, and handling state change. Summative worth 2%
Week 5 27 July	Event handling, Common gestures using gesture detector class	Chs.26-29	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on event listener and callback methods. Summative worth 2%



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 6 03 August	Fragments and managing overflow menus, Modern Android App Architecture	Chs.30-33 & 42	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on Fragments and menus. Summative worth 2% Assessment 2 Due (Individual Report)
Week 7 10 August	Floating Action Button, Snackbars and Tabbed Interface using TabLayout	Ch 45, 46	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on the Floating Action Button and Tab layout Summative worth 2% Assessment 4A Due (Group Formation)
Week 8 17 August	Recycler View and Card View Widget, Overview of Android Intents	Chs.47 - 49 & 52 - 55	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on displaying and responding to card item selections. Summative worth 2%
Week 9 24 August	Introduction to SQLite databases, Runtime permissions in Android	Chs.62 - 67	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on SQLite Database tutorials Summative worth 2% Assessment 3: Quiz
Week 10 31 August	Creating Audio/Video Recording and Ethical Considerations in Mobile App Development	Chs.68-72	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on cloud storage Summative worth 2%
Week 11 07 September	Working with Google maps API	Chs.73	Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on Android audio recording. Exercises in Handling different Android devices. Summative worth 2%



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
			Assessment 4B Due (Group assignment): Report Group Presentation (Cont.)
Week 12 14 September	Creating, Testing and Uploading Android App Bundle		Tutorial exercises based on lecture topics are intended to stimulate discussion. Exercises on working with Windows and iOS. Revision: Assessment 4B Due (Group assignment) Group Presentation
Week 13 21 September	Study review week and Final Exam Week		
Week 14 28 September	Examinations Continuing students - enrolments for T326 open	Please see exam timetable for exam date, time and location	
Week 15 06 October (Tue)	Student Vacation begins New students - enrolments for T326 open		
Week 16 12 October	• Results Released • Review of Grade Day for T226 – see Sections 2.6 and 3.2 below for relevant information. • Certification of Grades NOTE: More information about the dates will be provided at a later date through Moodle/KOI email.		
T326 26 October 2026			
Week 1 26 October	Week 1 of classes for T326		

2.5 Public Holiday Amendments

Please note: KOI is closed on all scheduled NSW Public Holidays.

There are no public holidays during the teaching weeks of this trimester.

2.6 Review of Grade, Deferred Exams & Supplementary Exams/Assessments

Review of Grade:



There may be instances when you believe that your final grade in a subject does not accurately reflect your performance against the marking criteria. Section 8 of the *Assessment and Assessment Appeals Policy* (www.koi.edu.au) describes the grounds on which you may apply for a Review of Grade.

If you have a concern about your marks and you are unable to resolve it with the Academic staff concerned, then you can apply for a formal Review of Grade as explained in section 3.2(e) Appeals Process below. Please note the time limits for requesting a review. Please ensure you read the Review of Grade information before submitting an application.

Review of Grade Day:

Final exam scripts will not normally be returned to students. Students can obtain feedback on their exam performance and their results for the whole subject at the Review of Grade Day. KOI will hold the Review of Grade Day for all subjects studied in T226. **The ROG day will be announced at a later date and the students will be notified through Moodle/KOI email.**

Only final exams and whole subject results will be discussed as all other assessments should have been reviewed during the trimester. Further information about Review of Grade Day will be available through Moodle.

If you fail one or more subjects and you wish to consider applying for a Review of Grade you are **STRONGLY ADVISED to** attend the Review of Grade Day. You will have the chance to discuss your final exam and subject result with your lecturer, and will be advised if you have valid reasons for applying for a Review of Grade (see Section 3.2 below and the *Assessment and Assessment Appeals Policy*).

A formal request for a review of grade may not be considered unless you first contact the subject coordinator to discuss the result.

Deferred Exams:

If you wish to apply for a deferred exam because you are unable to attend the scheduled exam, you should submit the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.

If you miss your mid-trimester or final exam there is no guarantee you will be offered a deferred exam.

You must apply within the stated timeframe and satisfy the conditions for approval to be offered a deferred exam (see Section 8.1 of the *Assessment and Assessment Appeals Policy and the Application for Assignment Extension or Deferred Exam Forms*). In assessing your request for a deferred exam, KOI will take into account the information you provide, the severity of the event or circumstance, your performance on other items of assessment in the subject, class attendance and your history of previous applications for special consideration.

Deferred mid-trimester exams will be held before the end of week 9. Deferred final exams will be held on two days during week 1 or 2 in the next trimester. You will not normally be granted a deferred exam on the grounds that you mistook the time, date or place of an examination, or that you have made arrangements to be elsewhere at that time; for example, have booked plane tickets.

If you are offered a deferred exam, but do not attend *you will be awarded 0 marks for the exam*. This may mean it becomes difficult for you to pass the subject. If you apply for a deferred exam within the required time frame and satisfy the conditions you will be advised by email (to your KOI student email address) of the time and date for the deferred exam. Please ensure that you are available to take the exam at this time.

Marks awarded for the deferred exam will be the marks awarded for that item of assessment towards your final mark in the subject.

Supplementary Assessments (Exams and Assessments):

A supplementary assessment may be offered to students to provide a final opportunity to demonstrate successful achievement of the learning outcomes of a subject. *Supplementary assessments are only*



offered at the discretion of the Board of Examiners. In considering whether or not to offer a supplementary assessment, KOI will take into account your performance on all the major assessment items in the subject, your attendance, participation and your history of any previous special considerations.

If you are offered a supplementary assessment, you will be advised by email to your *KOI student email address* of the time and due date for the supplementary assessment – supplementary exams will normally be held at the same time as deferred final exams during week 1 or week 2 of the next trimester.

You must pass the supplementary assessment to pass the subject. The maximum grade you can achieve in a subject based on a supplementary assessment is a PASS grade.

If you:

- are offered a supplementary assessment, but fail it;
- are offered a supplementary exam, but do not attend; or
- are offered a supplementary assessment but do not submit by the due date;

you will receive a FAIL grade for the subject.

Students are also eligible for a supplementary assessment for their final subject in a course where they fail the subject but have successfully completed all other subjects in the course. You must have completed all major assessment tasks for the subject and obtained a passing mark on at least one of the major assessment tasks to be eligible for a supplementary assessment.

If you believe you meet the criteria for a supplementary assessment for the final subject in your course, but have not received an offer, complete the *Complaint, Grievance, Appeal Form* and send your form to reception@koi.edu.au. The deadline for applying for supplementary assessment is the Friday of the first week of classes in the next trimester.

2.7 Teaching Methods/Strategies

Briefly described below are the teaching methods/strategies used in this subject:

- *Lectures* (2 hours/week) are conducted in seminar style and address the subject content, provide motivation and context and draw on the students' experience and preparatory reading.
- *Tutorials* (2 hours/week) include class discussion of case studies and research papers, practice sets and problem-solving and syndicate work on group projects. Tutorials often include group exercises and so contribute to the development of teamwork skills and cultural understanding. Tutorial participation is an essential component of the subject and contributes to the development of many of the graduate attributes (see section 2.2 above). Tutorial participation contributes towards the assessment in many subjects (see details in Section 3.1 for this subject). Supplementary tutorial material such as case studies, recommended readings, review questions etc. will be made available each week in Moodle.
- *Online* teaching resources include class materials, readings, model answers to assignments and exercises and discussion boards. All online materials for this subject as provided by KOI will be found in the Moodle page for this subject. Students should access Moodle regularly as material may be updated at any time during the trimester
- *Other contact* - academic staff may also contact students either via Moodle messaging, or via email to the email address provided to KOI on enrolment.

2.8 Student Assessment



Assessment is designed to encourage effective student learning and enable students to develop and demonstrate the skills and knowledge identified in the subject learning outcomes. Assessment tasks during the first half of the study period are usually intended to maximise the developmental function of assessment (formative assessment). These assessment tasks include weekly tutorial exercises (as indicated in the weekly planner) and low stakes graded assessment (as shown in the graded assessment table). The major assessment tasks where students demonstrate their knowledge and skills (summative assessment) generally occur later in the study period. These are the major graded assessment items shown in the graded assessment table.

Final grades are awarded by the Board of Examiners in accordance with KOI's Assessment and Assessment Appeals Policy. The definitions and guidelines for the awarding of final grades within the BIT degree are:

- HD High distinction (85-100%) an outstanding level of achievement in relation to the assessment process.
- DI Distinction (75-84%) a high level of achievement in relation to the assessment process.
- CR Credit (65-74%) a better than satisfactory level of achievement in relation to the assessment process.
- P Pass (50-64%) a satisfactory level of achievement in relation to the assessment process.
- F Fail (0-49%) an unsatisfactory level of achievement in relation to the assessment process.

Provided below is a schedule of formal assessment tasks and major examinations for the subject.

Assessment Type	When assessed	Weighting	Learning Outcomes Assessed
Assessment 1: Weekly Tutorial Exercises	Weeks 2 - 11	20%	a, b, c, d
Assessment 2: Individual Report	Week 6	25 %	a, b, d
Assessment 3: Quiz	Week 9	20%	a, b, c, d
Assessment 4: Project Group assignment	Report: Week 11 Presentation: Week 12	Report: 25% Individual Demonstration: 10%	b, c, d

Requirements to Pass the Subject:

To gain a pass or better in this subject, students must gain a *minimum of 50%* of the total available subject marks.

2.9 Prescribed and Recommended Readings

Provided below, in formal reference format, is a list of the prescribed and recommended readings.

Prescribed Text:

Smyth, N., 2025, Android Studio Meerkat Essentials - Java Edition, Payload Media, Meerkat Available from: ProQuest Ebook Central. ISBN: 9781965764145

Recommended Readings:

DiMarzio, J.F., 2017. *Beginning Android programming with Android Studio*. ISBN: 9781118705599

McDermid, D., 2015. *Ethics in ICT: An Australian perspective*. Frenchs Forest, NSW: Pearson Education. ISBN: 9781442552159

McDermid, D., 2008, *Ethics In ICT: An Australian perspective*. Frenchs Forest, NSW, Australia: Pearson Education. ISBN: 9780733993879

Smyth, N. 2020. *Android Studio 3.6 Development Essentials - Java Edition*. Payload Media, Cary. Available from: ProQuest Ebook Central. ISBN: 9781951442156

Zammetti, F., 2019, *Practical Flutter: Improve Your Mobile Development with Google's Latest Open-Source SDK*, Apress L. P., Berkeley, CA. Available from: ProQuest Ebook Central. ISBN: 9781484249734

Student Insights on Cybersecurity Careers:

<https://www.nist.gov/blogs/cybersecurity-insights/student-insights-cybersecurity-careers>

Ethics References:

- Australian Privacy Principles (APP): Reference the Office of the Australian Information Commissioner (OAIC) website. (<https://www.oaic.gov.au/privacy/australian-privacy-principles>)
- GDPR: Include the GDPR official regulation link (<https://www.dfat.gov.au/sites/default/files/nixora-group-eufta-submission.pdf>).
- ACSC Guidelines for data transfers (<https://www.cyber.gov.au/resources-business-and-government/essential-cybersecurity/ism/cyber-security-guidelines/guidelines-data-transfers>)
- ACSC Guidelines for software development (<https://www.cyber.gov.au/resources-business-and-government/essential-cybersecurity/ism/cyber-security-guidelines/guidelines-software-development>)
- The official Android Developer website (<https://developer.android.com/>)
- Android Studio Documentation: (<https://developer.android.com/studio/documentation>).
- Kotlin Documentation: (<https://kotlinlang.org/docs/home.html>)
- Android Developers YouTube Channel: The official Android Developers YouTube channel (<https://www.youtube.com/user/androiddevelopers>)

Security References:

- Australian Cyber Security Centre (ACSC): Official ACSC guidelines for implementing ICT security measures(<https://www.cyber.gov.au/resources-business-and-government/essential-cyber-security/ism/cyber-security-guidelines>).
- NIST Cybersecurity Framework: Reference the NIST website for guidelines (<https://www.nist.gov/cyberframework>).
- NIST SP 800-100, Information Security Handbook: A Guide for Managers (2024) (<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-100.pdf>)
- ISO/IEC 27001 standards
- NIST Cybersecurity Framework
- COBIT for Information Security
- General Data Protection Regulation (GDPR)



Journals:

- International Journal of e-Services and Mobile Applications
- International Journal of Mobile Human Computer Interaction
- Wireless Communications and Mobile Computing

Conference / Journal Articles:

Students are encouraged to read peer reviewed journal articles and conference papers. Google Scholar provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites.

3. Assessment Details

3.1 Details of Each Assessment Item

The assessments for this subject are described below. The description includes the type of assessment, its purpose, weighting, due date and submission requirements, the topic of the assessment, details of the task and detailed marking criteria, including a marking Rubrics.

Assessment submission is as per the instructions below. Please retain a photocopy and softcopy of all assessments. KOI expects students to submit their own original work in both assignments and exams, or the original work of their group in the case of group assignments.

Marking guides for assessments follow the assessment descriptions. Students should compare final drafts of their assessment against the marking guide before submission.

Assessment 1

Assessment type: Weekly tutorial exercises. Weeks 2 - 11

Assessment purpose: Students will be required to answer questions in weekly tutorial exercises based on the topics covered in lectures. This assessment contributes to learning outcomes a, b, c and d.

Value: 20% **Due Date:** Weekly (2% per week)

Assessment topic: The topics covered in each week.

Task details: Students must complete the weekly tutorial exercises and upload the answers to Moodle. Tutors will provide feedback to students during tutorial activities.

Assessment 2

Assessment Type: Individual assessment

Purpose: This assessment focuses on developing a multi-page Android application to track a Patient's activities, such as Blood Pressure Monitoring, exercise routines, suggesting diet plans, and hydration. Students will implement functionality to navigate between pages and pass data through intents while ensuring the privacy of sensitive user data such as personal health information. This assessment contributes to learning outcomes a, b and d.

Value: 25% **Due Date:** Week 6



Assessment topic: Blood Pressure Tracking App

Task details:

Overview: Develop a Blood Pressure Monitoring App. The App is used to assist Patients in monitoring their health/Blood pressure by logging and summarizing key small changes in lifestyle habits. The app will consist of Five primary pages, each building on the previous one and passing data forward using intents. This exercise will focus on the effective use of intents for navigation and secure data sharing while ensuring compliance with privacy regulations such as the General Data Protection Regulation (GDPR) and Australia's Privacy Principles (APPs). The reasons for storing the data and use of patients' data for research and for providing better solutions with the partners, e.g., Pharmaceutical companies, Electronics and Mechanical manufacturing companies and with Government agencies for improving services.

Objectives

- Intent Navigation: Demonstrate the use of intents for activity navigation and data sharing between multiple pages.
- Data Sharing and Privacy: Implement secure data sharing techniques using intents, ensuring sensitive user data is handled securely (e.g., health information). Students must also demonstrate how they comply with GDPR and Australia's Privacy Principles (APPs).
- User Interface: Design a user-friendly interface that encourages healthy habits and offers visual insights into the user's progress.
- Use of status/progress bar in each app and connecting the app with external website.

Pages and Features:

1. App Starting Page: Welcome screen with app name/logo and a "Start" button.
2. Profile Creating Page: User enters details such as Patient's name, date of birth (app calculates age), Picture, Registration date (using system date), weight, Doctors name and contact details and height.
3. Exercise Tracking Page: Users can select multiple exercises (running, walking, swimming, cycling, yoga, gym) and log duration and calories burned and suggest the goals.
4. Diet Tracking Page: Users log multiple meals (breakfast, lunch, dinner, snacks) with calories/Carb consumed and water intake along with dietician name and contact details.
5. Summary Page: Displays overall calories burned and consumed, net balance, exercise duration, and water intake with options to edit information and add motivational comments for next goal.
6. Attractive UI: Clean design, colourful icons, rounded buttons, smooth navigation, and engaging charts/graphs with progress bar on each page.
7. Use of compulsory Privacy Check box option in Starting Page with providing reasons for storing and sharing and dissemination of information with business partners for example the data for research, Privacy and Data Compliance reasons: Students must demonstrate how the app complies with GDPR (General Data Protection Regulation) and Australia's Privacy Principles (APPs), including consent management, secure data storage, and user rights to access/delete their data.
8. Connect the app with NSW Health website: <https://www.health.nsw.gov.au/>.



Technical Requirements:

- Use `Intent.putExtra()` to pass data between activities and retrieve it using `getIntent().getExtras()`.
- Ensure compliance with privacy principles (GDPR and APPs) when managing sensitive user data.
- Implement error handling for user inputs.

Deliverables:

1. Start with a title page including App title, Patients' details, and Registration date linked with system date.
2. Provide an introduction explaining the purpose of the app as a Monitoring with privacy compliance with Toast Message.
3. Describe the App features and data flow across pages, Sitemap, supported with a simple diagram.
4. Use code snippets only where necessary, focusing on intent passing and data handling with clear comments.
5. Include screenshots of Android Studio working app with time and date of each page showing user input, navigation, and the summary, with notes on privacy features such as consent or privacy notices.
6. Explain how the app complies with GDPR and APPs principles, including consent, right to access and delete data, secure storage, and minimal data collection.
7. Discuss the challenges encountered in applying privacy principles and how they were addressed.
8. Reflect on lessons learned, the importance of privacy, and how compliance improves user trust.
9. Conclude with a brief summary of the project and possible improvements.
10. Add references if external documents or frameworks such as GDPR or APPs guidelines were used.
11. Import the source file and upload it to the Moodle under the Individual assignment app source file section preferably in a zip folder.

Submission Requirements:

Students will prepare a comprehensive report including:

- An assignment cover page.
- A user manual describing how the application works, supplemented with Android studio with Java interface screenshots with time and date in the report in its related section
- Text referencing should be provided with at least 6-8 references.
- Before or after submission, all students should demonstrate and present their work and show to their tutor the functioning of the app on Android Studio.
- All documents and screenshots should be compiled into a single Microsoft Word format report.
- All source files must be uploaded on Moodle in a zipped file.

This assessment allows students to apply practical knowledge of Android development using intents, focusing on creating a seamless user experience while managing data across multiple activities. The context of creation of this application provides a relatable and engaging way for students to demonstrate their skills in a real-world application scenario.

Marking Rubric for Assessment 2: Individual assessment.

Value 25%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)



25% Functionality of the Application (using Intents)	App performs less than 50% of required tasks, intents are not correctly implemented.	App performs 50-64% of tasks, intents are used but with limited functionality.	App performs 65-74% of tasks, and intents are correctly used for basic navigation and data passing.	App performs 75-84% of tasks with effective use of intents for seamless navigation and data passing.	App performs more than 85% of tasks with advanced features, and intents are used seamlessly for data navigation.
25% Report Structure & Clarity	Report missing or poorly structured, no title page, introduction, or logical flow.	Basic structure with minimal sections (missing introduction, diagram, or conclusion)..	Adequate structure including title page, introduction, and conclusion; limited explanation of features or data flow..	Well-structured report including title page, introduction, diagram of data flow, and conclusion; clear explanation of features.	Exceptionally well-organized report with professional presentation, including title page, introduction, detailed data flow diagram, clear explanation of features, and strong conclusion..
25% Demonstration of App Features & Data Flow	Little to no explanation of app features; screenshots missing or irrelevant..	Basic description of app features with limited screenshots; weak explanation of data passing.	Clear description of app features with relevant screenshots; adequate explanation of navigation and data flow.	Strong explanation of app features supported with clear screenshots; data flow between pages explained logically.	Comprehensive explanation of app features with professional screenshots and detailed diagram, showing seamless data flow across all pages.
25% Privacy, Security & Reflection (GDPR/APPs Compliance)	No discussion of privacy, security, or challenges; GDPR/APPs not mentioned.	Basic mention of privacy or GDPR/APPs with little detail; minimal reflection on challenges.	Good discussion of privacy and GDPR/APPs principles; identifies some challenges and lessons learned.	Strong explanation of privacy compliance (consent, secure storage, user rights); clear reflection on challenges and solutions.	Exceptional analysis of privacy and GDPR/APPs compliance, including consent, secure storage, data rights; deep reflection on challenges, lessons learned, and how compliance builds user trust; references included.



Assessment 3

Assessment type: In-Class Quiz – individual assignment invigilated open book.

Purpose: This assessment will allow students to demonstrate their understanding of the topics discussed during tutorials. This assessment contributes to learning outcomes a, b, c and d.

Value: 20%

Due Date: Week 9 in usual tutorial timeslots

Submission requirements details: Quiz will be conducted **in class** and completed online via Moodle.

Task Details: The quiz will consist of a series of multiple-choice questions, short questions and match the subject content taught in weeks 1 – 8 inclusive

Marking Rubric for Assessment: 3

Value: 20%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Android Fundamentals and Setup 4 marks	Incorrect or missing identification of Android environment components. Unable to recognise basic app structure.	Correctly identifies basic Android setup steps and UI elements but with errors or omissions in explanations of architecture or layout usage.	Correctly explains and applies Android setup, UI layouts, and architecture in most questions with minor errors in terminology or structure.	Consistently correct explanation and application of Android setup, UI design, and architecture; minor or isolated minor errors only.	Accurately and consistently demonstrates full understanding of Android setup, UI design, layouts, and architecture with no conceptual errors.
UI, Layouts and View Binding 4 marks	Poor understanding of UI design, layouts, and view binding.	Basic recognition of UI components with limited application.	Good understanding of layouts and view binding with minor issues.	Very good understanding of UI design, layouts, and state handling.	Excellent and accurate application of UI design, layouts, and view binding.
Lifecycle, Events and Gestures 4 marks	Unable to explain lifecycle or event handling concepts.	Basic recall of lifecycle/events with limited understanding.	Adequate understanding with minor gaps in events or gestures.	Strong understanding of lifecycle, events, and gestures.	Expert understanding and accurate application of lifecycle and event handling.
Fragments and App Architecture 4 marks	No understanding of fragments, navigation, or architecture.	Basic awareness of fragments and navigation concepts.	Good understanding with minor errors in structure or navigation.	Very good understanding of fragments and modern architecture.	Excellent understanding and application of fragments, navigation, and architecture.
Applied Knowledge and Problem Solving 4 marks	Cannot apply concepts or interpret code/questions.	Limited application; mostly factual recall only.	Applies concepts with some minor errors.	Strong application and interpretation of scenarios/code.	Excellent problem solving and accurate interpretation across all question types.



Total Mark: / (20 marks)	COMMENTS:
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Assessment 4

Assessment Type: Project- group project.

Purpose: Students will develop an innovative, medium-sized application for a smartphone or tablet that can be commercialised. This will involve problem solving and design skills acquired in this subject. Students will be working in a group of 3-4 students. Each student is required to work towards this project. This assessment is designed to reinforce the foundation theories taught and developed students' skills and application of knowledge of the subject content. This assessment relates to learning outcomes b, c and d.

Value: 35%

Report – 25%

Individual Demonstration – 10%

Submission: Online on Moodle homepage.

Report: Due in Week 11, **Individual Demonstration and Group Presentation:** Due in Week 12

Task details: These days mobile is considered as a synonym for growth, and all businesses are now focusing on developing their mobile applications, The Complex nature and limited resources especially conducting eCommerce businesses for Service Industry (e.g.: Real estate agents, Train/Airline ticket booking, Hospitals/Medical centre, Hotel Booking, Internet/Mobile Service Provider) is crucial for continuity of business processed. A dedicated mobile application is easier to use and navigate and looks much better than a mobile website. In this assignment, students will be developing a basic Ecommerce Android application. First, share the business name/idea you selected for application development with your tutor, and once the tutor approves it, you can start working on the assignment. This assessment has two parts.

Students will complete the development of an entire eCommerce application.

The application should have the following Pages.

1. Registration Page (Customers should be able to register themselves) for the Service Provider.
2. Login (Registered customers can log in to the application. Log-in fields should include the username or email and password) with compulsory Privacy check box option and Toast Message.
3. Home Page (The homepage can be in a card design, with each card representing a product/service and its price. The page should scroll to allow for additional cards that are arranged and represent all the products. From this home screen, the user should be able to access a settings menu and product search bar. They should also be able to access their product cart)
4. Detail page: Once a user clicks on any product on the home page, it should navigate to the product detail page. Users can select the quantity of a product/service and add the product/service to the cart;
5. Search (Users should be able to search for products and information while using the app. The search bar should be typed ahead, allowing the user to get recommendations while they type and select the options that appear)
6. Cart (Users need the ability to see the products they have in their cart quickly. When the user selects the cart, they should land on a page that shows all of the products they have placed in the cart). The information they should see includes: The user should also see the price of the product and be able to update the quantity of each product they have in the cart. The user should also see the total cost of all the products they have in the cart, and they can finalise.

7. Checkout (When a user enters a secure checkout, they should have the option to enter their personal information. Fields include Full Name, Address, Phone and Email)
8. Each Page should have a progress/ status bar.
9. Purchase History (From the menu, customers should be able to see their order history. Users want to see what they have ordered in the past. The order history page should show all of their orders for the past six months)
10. Shipping Information page using Intent, should fill the data from the Checkout page.

This e-commerce Android application should be developed using Android Studio, XML, and Java only.

Each group will provide a group work report on how the group has worked together to produce the final report. It should contain the demonstration of the main deliverable itself in the final report. Each group documents how members discuss and division of responsibilities, challenges faced and describes how the individual efforts capitalised on the strengths and weaknesses of each group member. It will be used as evidence of individual contributions to the group. It is, therefore, in each member's interest to ensure that their contribution to the final report is complete.

Each group is expected to have at least 4 group meetings for the group assignment. The minutes of group meetings should be documented and attached as an appendix to this group work report, clearly indicating who was present, issues and actions, agreed timelines, and the like. The group work report must demonstrate a fair and reasonable distribution of work amongst group members. Periodic information, such as emails or diary entries, must be inserted into the correct section in chronological order. In week 12, after submission, all students and group members should demonstrate and present their piece of work and show to their tutor the functioning of the app on Android Studio.

If the submitted group work report suggests that not all contributions were of equivalent standard and effort, differential marks will be awarded to individuals within the same group. It must also document what individual group members understood as their allocated tasks and that particular group members submitted assigned work of acceptable standard and quality by the agreed-upon date.

Submission requirements details: Students will also prepare a report. Create a user manual report to show how your application works. Also, you need to provide screenshots for the user interface. Assignment cover, user manual and screenshots should be submitted in a single Word-format report. All source files must be uploaded to the Moodle as a zip file.

Marking Rubric for Assessment 4: Individual contributed.

Value 10%

Criteria	Unsatisfactory (0 – 49%)	Satisfactory (50 – 64%)	Effective (65 – 74%)	Excellent (75 – 84%)	Exceptional (85 – 100%)	Weight
Demonstration of Personal Contribution	Demonstrates little to no personal contribution or understanding of the task.	Demonstrates basic personal contribution but lacks depth in understanding or application.	Demonstrates a clear personal contribution with a solid understanding and application.	Demonstrates a strong personal contribution with thorough understanding and application.	Demonstrates an outstanding personal contribution with exceptional understanding and application.	3%
Understanding of Application Features	Shows minimal understanding of application features and functionality.	Shows adequate understanding of application features but with limited details.	Shows good understanding of application features with some detailed insights.	Shows excellent understanding of application features with comprehensive details.	Shows exceptional understanding of application features with in-depth insights.	2%
Ability to Navigate and Explain the App	Has significant difficulty in navigating the app and	Can navigate the app but explains features with some difficulty.	Navigates the app well and explains features effectively.	Navigates the app smoothly and explains features with high clarity.	Navigates the app expertly and explains features with exceptional clarity and insight.	2.5%



Criteria	Unsatisfactory (0 – 49%)	Satisfactory (50 – 64%)	Effective (65 – 74%)	Excellent (75 – 84%)	Exceptional (85 – 100%)	Weight
	explaining its features.					
Response to Questions	Provides unsatisfactory responses to questions with a lack of understanding.	Provides basic responses to questions, demonstrating a moderate understanding.	Provides good responses that demonstrate a clear understanding of the project.	Provides very good, well-thought-out responses demonstrating a thorough understanding.	Provides exceptional responses with a deep and comprehensive understanding.	2.5%

Marking Rubric for Assessment 4: Group Report Rubric.

Value 25%

Criteria	Unsatisfactory	Satisfactory	Effective	Excellent	Exceptional	25%
	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)	
Functionality, of the application	Performs less than 50% of tasks.	Performs 50%-64% of tasks.	Performs 65%-74% of tasks.	Performs 75%-84% of tasks.	Performs more than 85% of task	6
Sophistication of the application, quality of the design	Very difficult to read, structure not clear.	Some difficulty in reading, structure lacking in some parts	Well written, but the structure is not very clear.	Well written and well structured.	Quality is excellent. Very well clearly written and structured.	5
Use of Intent, Data Sharing & Consent, GDPR Guidelines	No evidence of privacy consideration; intent unclear; personal data collected indiscriminately; consent absent or misleading; clear GDPR breaches.	Basic mention of privacy; intent stated but vague; collects more data than needed; consent is generic "tick-box" with gaps; partial GDPR compliance.	Clear intent statement; data minimisation mostly observed; informed consent obtained in standard form; GDPR duties (lawful basis, user rights) largely covered, with minor omissions.	Comprehensive, transparent data-flow diagram; granular, opt-in consent matching each purpose; strong data-minimisation; demonstrates full GDPR compliance incl. retention policy and user-rights workflow.	Privacy-by-design embedded throughout; just-in-time contextual consent; dynamic data-sharing controls; documented DPIA and risk mitigations; automated user-rights handling (access, erasure, portability); sets best-practice benchmark.	4
Innovation of the application, and use of researched techniques	No support for the position taken in the answer.	Provides some support for the position taken.	Provides a good level of support for positions taken.	Provides an excellent level of support for positions taken.	Provides an exceptional level of support for positions taken.	4
Communication of design in professional report and	No conclusion presented	Conclusion based on some argument	Conclusion based on sound argument	Conclusion based on an extensive argument	Conclusion based on an extensive and compelling argument	3



source code comments						
Report on identification of requirements, researched solutions and priorities	Very little subject matter knowledge shown.	Adequate subject matter knowledge shown.	Sound subject matter knowledge shown.	High level of subject matter knowledge shown.	High level of subject matter knowledge and synthesis with theory shown	3

3.2 General information about assessment

a) Late Penalties and Extensions

An important part of business life and key to achieving KOI's graduate outcome of Professional Skills is the ability to manage workloads and meet deadlines. Completing assessment tasks on time is a good way to master these habits.

Students who miss mid-trimester tests and final exams without a valid and accepted reason may not be granted a deferred exam and will be awarded 0 marks for the assessment item. Assessment items which are missed or submitted after the due date/time will attract a penalty unless there is a compelling reason (see below). These penalties are designed to encourage students to develop good time management practices, and to create equity for all students.

Any penalties applied will only be up to the maximum marks available for the specific piece of assessment attracting the penalty.

Late penalties, granting of extensions and deferred exams are based on the following:

In Class Tests and Quizzes (excluding Mid-Trimester Tests)

- Generally, extensions are not permitted. A make-up test may only be permitted under very special circumstances where acceptable supporting evidence of illness, hardship or unavoidable problems preventing completion of the assessment is provided (see section (b) below). The procedures and timing to apply for a make-up test (only if available) are as shown in the section *Applying for an Extension* (see below).
- Missing a class test will result in 0 marks for that assessment item unless the above applies.

Written Assessments and Video Assessments

- There is a late penalty of *5% of the total available marks* per calendar day unless an extension is approved (see *Applying for an Extension* section below).

Presentations

- Generally, extensions are not permitted. Missing a presentation will result in 0 marks for that assessment item. The rules for make-up presentations are the same as for missing in-class tests (described above).

For group presentations, if serious circumstances prevent some members of the group from participating, the members of the group who are present should make their contributions as agreed. If a make-up presentation is approved, the other members of the group will be able to make their individual presentation later and will be marked according to the marking rubric. A video presentation may be used to facilitate the process.

Mid-Trimester Tests and Final Exams



If students are unable to attend mid-trimester tests or final exams due to illness, hardship or some other unavoidable problem (acceptable to KOI), they must:

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days after the exam date*.
- Provide acceptable documentary evidence (see section (b) below).
- Agree to attend the deferred exam as set by KOI if a deferred exam is approved.

Deferred exam

- There will only be one deferred exam offered.
- Marks obtained for the deferred exam will be the marks awarded for that assessment item.
- If you miss the deferred exam you will be awarded 0 marks for the assessment item. This may mean you are unable to pass the subject.

b) Applying for an Extension

If students are unable to submit or attend an assessment when due, they must

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.
- Provide acceptable documentary evidence in the form of a medical certificate, police report or some other appropriate evidence of illness or hardship, or a technician's report on problems with computer or communications technology, or a signed and witnessed statutory declaration explaining the circumstances.
- Students and lecturers / tutors will be advised of the outcome of the extension request as soon as practicable.

Please remember there is no guarantee of an extension being granted, and poor organisation is not a satisfactory reason to be granted an extension.

c) Referencing and Plagiarism

Please remember that all sources used in assessment tasks must be suitably referenced.

Failure to acknowledge sources is plagiarism, and as such is a very serious academic issue. Students plagiarising run the risk of severe penalties ranging from a reduction in marks through to 0 marks for a first offence for a single assessment task, to exclusion from KOI in the most serious repeat cases. Exclusion has serious visa implications. The easiest way to avoid plagiarising is to reference all sources.

Harvard referencing is the required method – in-text referencing using Author's Surname (family name) and year of publication. A Referencing Guide, "Harvard Referencing", and a Referencing Tutorial can be found on the right-hand menu strip in Moodle on all subject pages.

An effective way to reference correctly is to use *Microsoft Word's* referencing function (please note that other versions and programs are likely to be different). To use the referencing function, click on the References Tab in the menu ribbon – students should choose *Harvard*.

Authorship is also an issue under plagiarism – KOI expects students to submit their own original work in both assessment and exams, or the original work of their group in the case of a group project. All students agree to a statement of authorship when submitting assessments online via Moodle, stating that the work submitted is their own original work.

The following are examples of academic misconduct and can attract severe penalties:



- Handing in work created by someone else (without acknowledgement), whether copied from another student, written by someone else, or from any published or electronic source, is fraud, and falls under the general Plagiarism guidelines.
- Copying / cheating in tests and exams is academic misconduct. Such incidents will be treated just as seriously as other forms of plagiarism.
- Students who willingly allow another student to copy their work in any assessment may be considered to assist in copying/cheating, and similar penalties may be applied.

Where a subject coordinator considers that a student might have engaged in academic misconduct, KOI may require the student to undertake an additional oral exam as a part of the assessment for the subject, as a way of testing the student's understanding of their work.

Further information can be found on the KOI website.

d) Reasonable Adjustment

The Commonwealth Disability Discrimination Act (1992) makes it unlawful to treat people with a disability less fairly than people without a disability. In the context of this subject, the principle of Reasonable Adjustment is applied to ensure that participants with a disability have equitable access to all aspects of the learning for the subject. For assessment, this means that barriers to their demonstrating competence are removed wherever it is reasonably practical to do so.

Examples of reasonable adjustment in assessment may include:

- provision of an oral assessment, rather than a written assessment
- provision of extra time
- use of adaptive technology.

The focus of the adjusted assessment should be on enabling the student to demonstrate achievement of the learning outcomes for the subject, rather than on the method of assessment.

e) Appeals Process

Full details of the KOI *Assessment and Assessment Appeals Policy* may be obtained in hard copy from the Library, and on the KOI website www.koi.edu.au under *Policies and Forms*.

Assessments and Mid-Trimester Exams:

Where students are not satisfied with the results of an assessment, including mid-trimester exams, they have the right to appeal. The process is as follows:

- Discuss the assessment with their tutor or lecturer – students should identify where they feel more marks should have been awarded – students should provide valid reasons based on the marking guide provided for the assessment. Reasons such as “*I worked really hard*” are not considered valid.
- If still not satisfied, students should complete an Application for Review of Assessment Marks form, clearly explaining the reasons for seeking a review. This form is available from the KOI website under *Policies and Forms* and is also available at KOI Reception (Level 9, 11 York St). The completed Application for Review of Assessment Marks form should be submitted as explained on the form with supporting evidence attached to academic@koi.edu.au.
- The form must be submitted within *ten (10) working days* of the return of the marked assessment, or within *five (5) working days* after the return of the assessment if the assessment is returned after the end of the trimester.



Review of Grade – whole of subject and final exams:

Where students are not satisfied with the results of the whole subject or with their final exam results, they have the right to request a Review of Grade – see the *Assessment and Assessment Appeals Policy* for more information.

An *Application for Review of Grade/Assessment Form* (available from the KOI Website under *Policies and Forms* and from KOI Reception at Level 9, 11 York St) should be completed clearly explaining the grounds for the application. The completed application should be submitted as explained on the form, with supporting evidence attached to academic@koi.edu.au.